

Program Assignment 3 (Spring 2026)

Due Date: **June 09, 2026**

[The following is adapted from wiki ([National Resident Matching Program - Wikipedia](#))]

The National Resident Matching Program (NRMP) (<https://www.nrmp.org/>) is a U.S.-based private, non-profit, and non-governmental organization established in 1952. Its mission is to match U.S. medical school students and graduates with residency and fellowship training programs at teaching hospitals across the United States.

The matching process is a generalization of the **stable marriage problem**, and the algorithm used shares many similarities with that model. A simplified version of the NRMP's matching algorithm is available on their website (<https://www.nrmp.org/intro-to-the-match/how-matching-algorithm-works/>). However, this overview omits complexities such as handling couples (pairs of applicants who wish to match together), second-year positions, and unfilled residency slots. The full algorithm, including these features, is detailed in the following academic reference:

Roth, and Peranson (September 1999). "The Redesign of the Matching Market for American Physicians: Some Engineering Aspects of Economic Design". The American Economic Review. 89 (4): 748–780. doi:10.1257/aer.89.4.748. PMID 29115787.

Assignment Description

Please design and implement an information system to assist with resource matching and allocation, with the goal of achieving optimal matching efficiency. Drawing inspiration from the NRMP and the stable marriage problem, the system should simulate a real-world matching scenario, such as applicants matching with institutions, students with dormitories, or job seekers with job openings. Your system design must include the following four components:

1. **Environment Description:** Describe the context in which your system operates. Clearly define the types of resources being matched (e.g., applicants and programs, tasks and machines, students and supervisors). Include the structure of the environment, relevant constraints (e.g., preferences, capacities), and any assumptions made. You should also discuss user roles (e.g., applicants, providers) and their interaction with the system.
2. **System Design:** Present the architecture of your proposed system. Describe how users input their data (e.g., preferences, qualifications, capacities), how the system processes this data, and how results are displayed. Consider user interfaces (web,

console, or mobile), data handling methods (e.g., preference lists, priority scores), and system components (e.g., matching engine, data validator, output formatter). You may include diagrams, UI mockups, or data flow representations to illustrate your design

3. **Problem Definition:** Define the core problems your system is intended to solve.

These may include:

- Handling mismatches between resource availability and demand
- Managing incomplete or inconsistent preference data
- Ensuring fairness and stability in matching
- Dealing with special cases such as ties, priorities, or constraints

4. **Algorithm Design:** Design and describe the algorithms used in your system. Your algorithm should address the problems defined in the previous section. You may use or adapt well-known algorithms such as:

- Gale–Shapley stable marriage algorithm
- Maximum bipartite matching
- Greedy heuristics or optimization-based models

At least one algorithm must be fully implemented and included in your report. Provide pseudocode, complexity analysis, and test results. Discuss how the algorithm ensures stable and optimal matching based on your system's goals.

Deliverables (Due Date: 13:10, June 09, 2026)

- A **hardcopy report** detailing your system design, problem definitions, and algorithm implementations.
- An **oral presentation** and **system demonstration** to introduce your work to the class.

Grading Policy

Baseline: Full credit for meeting the minimum requirements, including a clear description of the system environment, system design, problem definitions, at least one correctly implemented algorithm, and completion of both the written report and oral presentation.

Bonus: Additional credit will be awarded for:

- Defining more relevant and insightful problems in the system.

- Proposing a wider range of appropriate and creative algorithms.
- Implementing more algorithms correctly and effectively.

Grading Distribution

- Written Report: 50%
- Presentation & System Demonstration: 50%

Report Format

The report should be between **20** and **30** pages in length, formatted on **A4-size paper**. Please follow the formatting and structural guidelines below:

A. Cover Page

Include the following information clearly:

- Course title and section
- Student name
- Student ID number

B. Report Body

- Use **1.5 line spacing** throughout the report.
- Font size should be **12 point** for the main text and **14 point** for section titles.
- Each page must include a **page number**.

The body of the report should include the following sections:

1. Context, System, and Problem Definition

Describe the environment, the proposed system, and the specific problems the system aims to solve.

2. Proposed Algorithms

Detail the algorithms designed to address the defined problems, including any assumptions and design considerations.

3. Implementations and Results

Present the implemented algorithm(s), describe how they were tested, and provide the results.

4. Discussion

Analyze the strengths and limitations of your system and algorithms. Reflect on challenges encountered and possible future improvements.

Other Notes:

- Plagiarism will result in a score of zero.
- Please conduct your oral presentations in the spirit of collaborative learning, sharing, and mutual improvement. Learn from each other by exchanging insights and reflections.
- Please submit your assignments on time; **late submissions will not be accepted for credit.**